MILITARY SPECIFICATION

SABOT, CAP SEAL AND CAPS FOR 40MM, XM576E2

1. SCOPE

1.1 This specification covers Sabot, Cap Seal and Caps for 40MM, XM576E2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1310

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2.1 (Cont'd)

DRAWINGS

Munitions Command

B10542604	- Sabot
B10542606	- Cap, Female
B10542499	- Cap, Male
A10542603	- Seal, Cap
IEL 8648429	- Index of Inspection Equipment Lists for Sabot, Cap Seal and Caps for 40MM, XM576E2

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Sabot.

3.2.1 Tensile.-The sabot shall withstand a force of 1050 pounds minimum when pulled apart.

3.2.2 Water immersion.-The sabot shall sink below the surface when immersed in a container of water at 70 ± 5 degrees farenheit.

3.3 Plastic defects.-All components shall be free from blisters, bubbles, cracks, crazing, mottle, porosity or any other defects.

3.4 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Burr.-No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.4.2 Foreign matter.-No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

2 Sets of each component representing each mold cavity

The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacture processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Lot.

4.3.1.1 Lot formation.-A lot shall consist of Sabots, Caps and Cap Seals produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.2 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with standard MIL-STD-1235, may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examinations.

4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged at the beginning of each 8 hours production from each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the components dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool. All material produced prior to the periodic inspection will be rejected and screened whenever a defected tool control dimension is found.

4.3.2.2 Cap, Female, Drawing B10542606.

Defect	Method of Inspection
CRITICAL	
None defined	

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Diameter of flange	Gage
102.	Material defective	Visual
MINOR		
201.	Diameter forward or flange	Gage
202.	Depth of seal groove	Gage
203.	Length from seal groove end to diameter of flange	Gage
204.	Location of seal groove	Gage
205.	Thickness through seal groove web	Gage
206.	Radii missing or incorrect	Visual (see 6.1)
207.	Burr	Visual
208.	Foreign matter	Visual
209.	Marking missing	Visual

4.3.2.3 Cap, Male, Drawing B10542499.

CRITICAL		
	None defined	
MAJOR		
101.	Diameter of flange	Gage
102.	Material defective	Visual
MINOR		
201.	Diameter forward of flange	Gage
202.	Length of lip seal	Gage
203.	Length from lip seal end to diameter of flange	Gage
204.	Location of lip seal	Gage
205.	Width of lip seal	Gage
206.	Thickness through lip seal web	Gage
207.	Radii missing or incorrect	Visual (see 6.1)
208.	Burr	Visual
209.	Foreign matter	Visual
210.	Marking missing	Visual

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4.3.2.4 Seal, Cap, Drawing A10542603.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Diameter	Gage
202. Length, total	Gage
203. Foreign matter	Visual

4.3.2.5 Sabot, Drawing B10542604.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of diameter of forward cavity with diameter of flange	Gage
102. Concentricity of rear outside diameter with diameter of flange	Gage
103. Diameter of flange	Gage
104. Diameter at rear (outside)	Gage
105. Diameter of forward cavity	Gage
106. Depth of forward cavity	Gage
107. Thickness of flange	Gage
108. Center hole missing	Visual
109. Material defective	Visual
MINOR	
201. Length, total	Gage
202. Radii missing or incorrect	Visual (see 6.1)
203. Burr	Visual
204. Foreign matter	Visual
205. Marking missing	Visual

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with 4.4.

4.3.4 Sabot.

4.3.4.1 Tension.-Ten sabots from each lot of 10,000 sabots or fraction thereof shall be subjected to this test. Failure of two or more sabots to comply with the requirements of 3.2.1 shall be cause for rejection of the lot. In the event that only one sabot fails to comply with the requirements, another sample of twenty sabots shall be selected from the same lot and subjected to the same test. If one sabot in this second sample fails to comply with the requirements, the lot shall then be rejected.

4.3.4.2 Water immersion.-Ten sabots from each lot of 10,000 sabots or fraction thereof shall be subjected to this test. Failure of two or more sabots to comply with the requirements of 3.2.2 shall be cause for rejection of the lot. In the event that only one sabot fails to comply with the requirement, another sample of twenty sabots shall be selected from the same lot and subjected to the same test. If one sabot in the second sample fails to comply with the requirements, the lot shall then be rejected.

4.3.5 Inspection equipment.-Index of Inspection Equipment List No. IEL-8648429 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation of the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures.

4.4.1 Sabot.

4.4.1.1 Tension.-From the samples supplied in accordance with 4.3.4.1, each sabot will be center drilled to a 1.1353 $\pm$ .0005 diameter hole. The test shall be performed using equipment capable of performing the test accurately and consistently. The sabot shall be mounted in an approved fixture. In performing the test, a force shall be applied in opposite directions at a speed of one inch per minute and the maximum force required to pull the sabot apart shall be recorded (see Figure I). The equipment shall be calibrated prior to the start of each series of tests.

4.4.1.2 Water immersion.-The samples supplied in accordance with 4.3.4.2 shall be submerged in a container of clean water approximately one foot in diameter and one foot deep. Care should be taken to assure that no foreign matter or air bubbles are entrapped on the surfaces of the sabots. While under water, the sabots shall be released of any external forces so that the sabots may either sink to the bottom of the container or rise to the surface.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level A.-Not applicable

5.1.2 Level B.-Not applicable

5.1.3 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

5.2 Data card.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

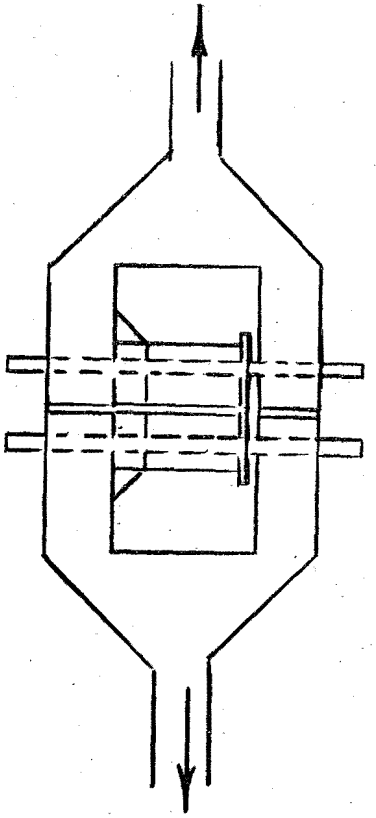
Army - MU

Preparing activity

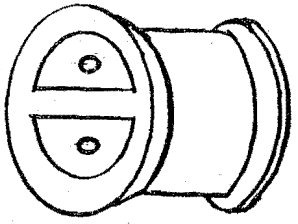
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TENSILE FIXTURE (COMPLETE)



SABOT (CORED) WITH HALF-MOON INSERTS FOR TENSILE FIXTURE

(FIG. 1)

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